Exploratory Data Analysis

Summary of Key EDA Findings

- Overall default rate is **39.1%**, implying a moderate data imbalance, which should be addressed by applying SMOTE.
- **Unemployed and Student users** have the highest default rates (~49%), nearly double that of Self-Employed users (~26%).
- Users with **lower credit scores** (300–550 range) have higher default probability.
- The default users have more **missed payments and repayment delays**.
- **Higher debt-to-income ratio** leads to higher default probability.
- `monthly_income` and `credit_score` exhibit a very high correlation (0.96), indicating multicollinearity; therefore, `monthly_income` is removed in the preprocessing stage.
- `customer_segment` and `risk_score` are removed in the preprocessing stage due to potential data leakage concerns.

Dataset Overview

Table 1: Dataset Overview

Item	Value
Total Records	10,345
Total Variables	17
Defaulted Users	4,040
Non-Defaulted Users	6,305
Default Rate	39.1%
Missing Values	None

Interpretation: The dataset contains 10,345 transactions across 17 variables. Overall, 39.1% of users defaulted on their repayments.

Descriptive Statistics of Numerical Variables

Table 2: Descriptive Statistics of Numerical Variables

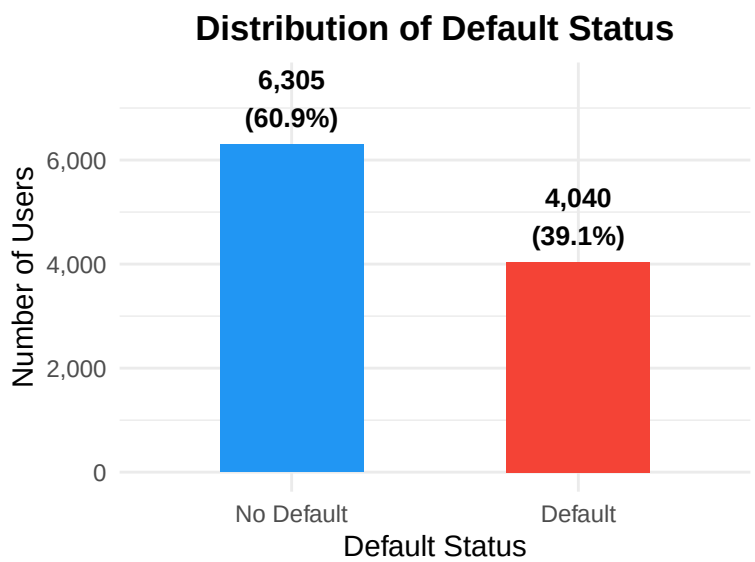
Variable	Mean	Median	SD	Min	Max
age	38.56	39.00	12.13	18	59.00
monthly_income	35053.38	23800.52	27084.52	5000	145767.12
credit_score	448.18	403.00	136.52	300	850.00
purchase_amount	3979.40	5000.00	1500.83	100	5000.00
repayment_delay_days	8.74	9.00	6.78	0	33.00

missed_payments	1.02	1.00	1.00	0	7.00
debt_to_income_ratio	0.18	0.14	0.12	0	0.72
risk_score	198.53	202.80	67.54	0	398.00
app_usage_frequency	5.52	5.55	2.59	1	10.00

Interpretation:

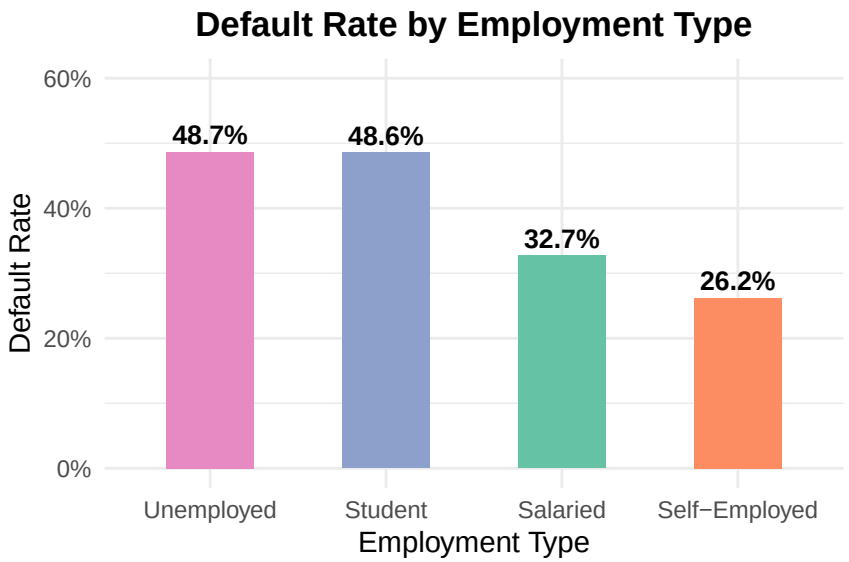
- **Monthly Income** ranges widely, with extreme high values identified as outliers that warrant further attention in preprocessing.
- **Credit Score** ranges from 300 to 850, with a median of 403, suggesting many users have low creditworthiness.
- **Repayment Delay Days** averages 8.7 days, with some users delaying up to 33 days. **Missed Payments** ranges up to 7, and **Debt-to-Income Ratio** reaches 0.72. The extreme values in these three variables are removed in the preprocessing stage.

Target Variable Distribution



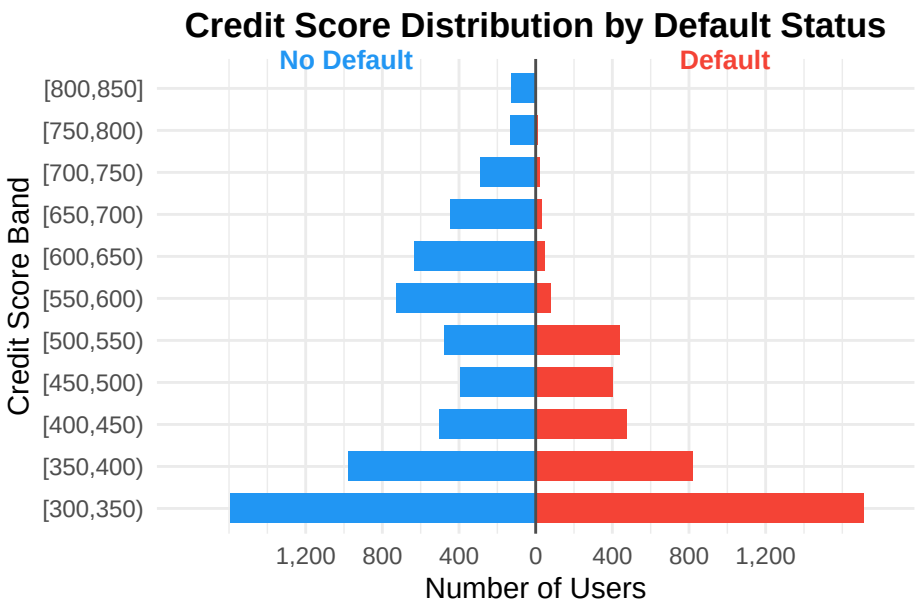
Interpretation: The dataset has a moderate data imbalance problem. 61% of users did not default, while 39% did. SMOTE will be applied in data preprocessing stage to deal with this problem.

Default Rate by Employment Type



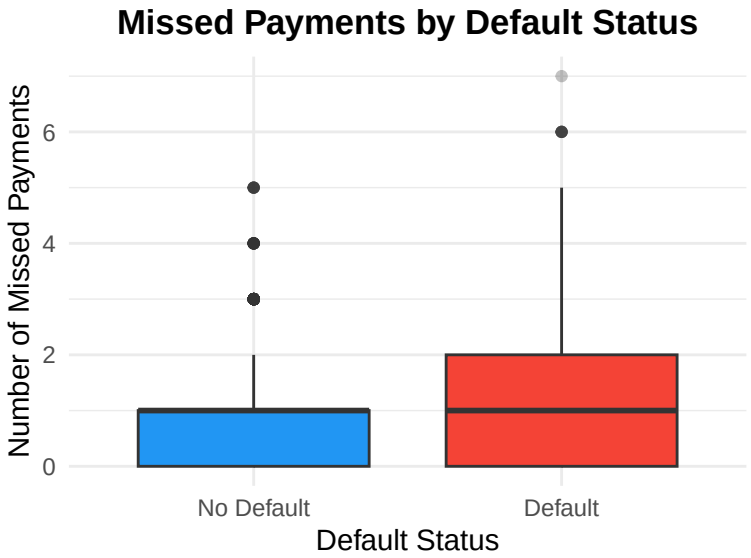
Interpretation: Unemployed users (48.7%) and Students (48.6%) have nearly double the default rate of Self-Employed users (26.2%) and significantly higher than Salaried workers (32.7%). This is because users with unstable or no income face greater difficulty in paying installment punctually.

Credit Score Distribution by Default Status

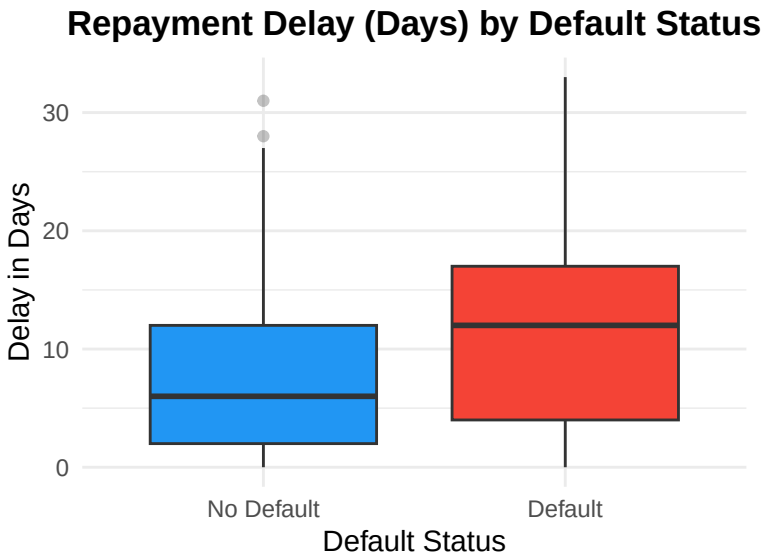


Interpretation: Users who defaulted tend to have a low credit score, ranging from 300 to 550, while non-defaulters show a wider and slightly higher distribution. This means that lower credit scores are associated with higher default risk.

Missed Payments & Repayment Delay vs. Default

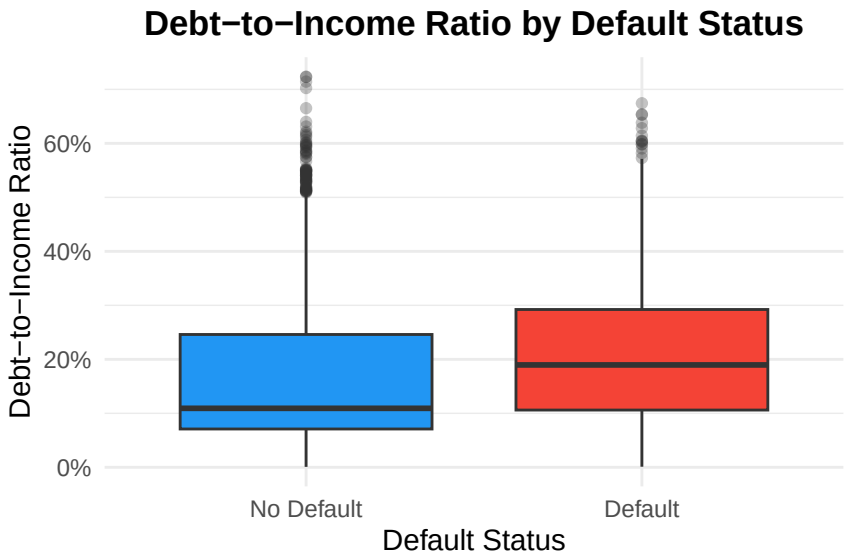


Interpretation: Defaulters have a wider distribution and a higher upper quartile of missed payments compared to non-defaulters. This is a strong signal of default risk, indicating users with repeatedly miss payments (e.g., 2 or more) are more likely to default.



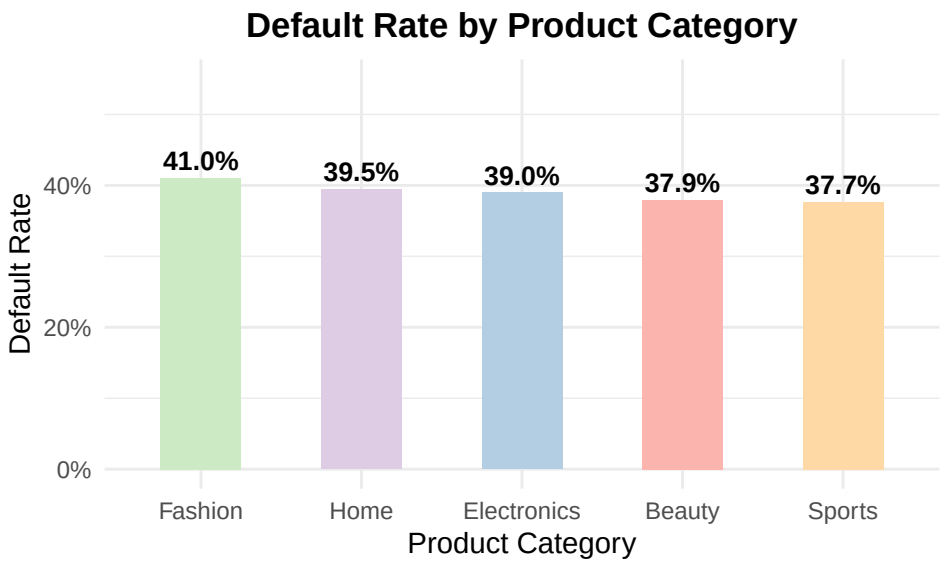
Interpretation: Defaulters also tend to delay repayments significantly longer than non-defaulters. The higher median repayment delay days among defaulters suggests that late payment behaviour is also a good predictor.

Debt-to-Income Ratio vs. Default



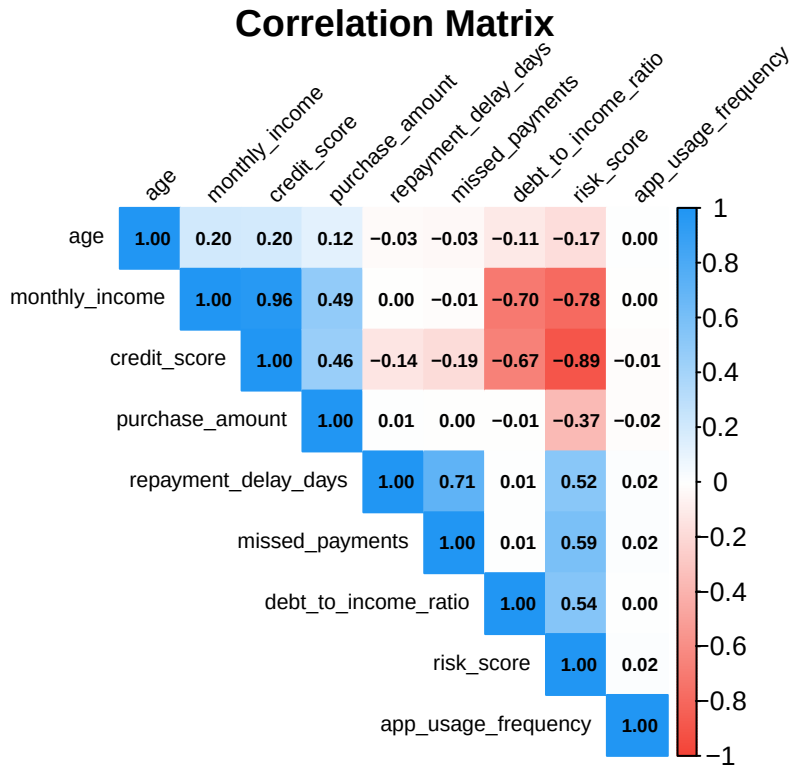
Interpretation: Defaulters have a significantly higher median debt-to-income ratio compared to non-defaulters. Users spending a larger proportion of their income on debt repayments are correlated with higher financial risk.

Default Rate by Product Category



Interpretation: Default rates across product categories are relatively similar, ranging from 37.7% (Sports) to 41.0% (Fashion). While Fashion purchases show a slightly higher default rate, the differences are not dramatic.

Correlation Heatmap of Numerical Variables



Interpretation: The correlation matrix reveals several notable relationships.

- **`repayment_delay_days`** is positively correlated with **`missed_payments`**, meaning that the longer a user delays a payment, the more likely the transaction will be flagged as a missed payment.
- **`credit_score`** is negatively correlated with **`debt_to_income_ratio`**, suggesting that users with heavier debt burdens are perceived as higher risk.
- Notably, **`monthly_income`** and **`credit_score`** have an extremely high correlation, indicating severe multicollinearity; **`monthly_income`** is therefore removed from modelling.
- Furthermore, **`risk_score`** shows abnormally strong correlations with multiple variables (e.g., r = -0.89 with **`credit_score`**), which raises potential data leakage concerns and should be removed.